

Interval-Censored Survival Analysis of Prostate Cancer Remission Data

Abstract

This project investigates remission times among prostate cancer patients using interval-censored survival data. A Weibull maximum likelihood estimation procedure was applied to estimate the shape and scale parameters of the survival distribution. Parametric survival estimates were compared with nonparametric survivor functions, and probability-probability (P-P) plots were used to assess model adequacy. Additional analyses examined the impact of treating interval-censored observations as right-censored and evaluated model performance under reduced sample sizes. Results suggest that the Weibull model provides a reasonable fit for the full dataset, while model adequacy deteriorates when only a subset of the observations is available. The study also demonstrates that simplifying interval-censored observations into exact event times may substantially alter survival estimates and lead to overestimation of survival times.

Methodology

The analysis was conducted using survival analysis techniques for interval-censored data. A Weibull distribution was fitted using maximum likelihood estimation, and nonparametric survivor functions were obtained using the `survfit()` function. Model adequacy was assessed using P-P plots.

Weibull Maximum Likelihood Estimation

The MLEs are calculated as,

Shape parameter = 2.004489

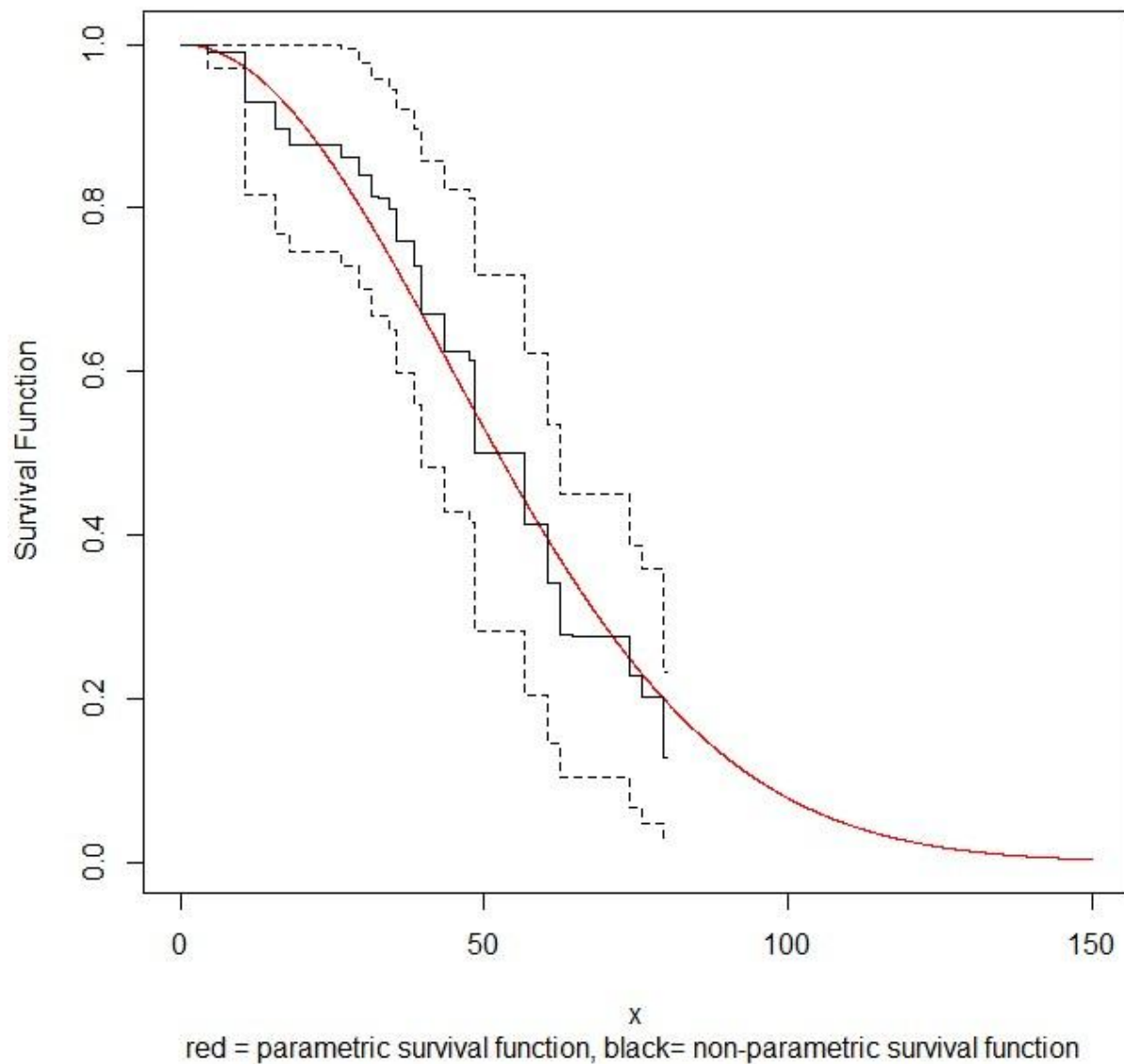
Scale parameter = 62.71555

We also find the confidence intervals,

time	n.risk	n.event	survival	std.err	lower	95% CI	upper	95% CI
4.5	100.0	1.00e+00	0.990	0.00996		0.9703		1.000
10.5	98.0	6.00e+00	0.929	0.05421		0.8151		1.000
11.5	91.0	7.33e-15	0.929	0.05421		0.8151		1.000
13.0	91.0	3.87e-10	0.929	0.05421		0.8151		1.000
15.5	91.0	3.31e+00	0.896	0.05834		0.7679		1.000
18.0	84.7	1.69e+00	0.878	0.05854		0.7470		1.000
20.0	82.0	6.20e-15	0.878	0.05854		0.7470		1.000
26.5	77.0	1.43e+00	0.861	0.05859		0.7281		0.995
29.5	71.6	1.82e+00	0.840	0.05921		0.7013		0.978
31.5	68.8	2.17e+00	0.813	0.06020		0.6679		0.958
32.5	65.6	2.56e-02	0.813	0.06018		0.6675		0.958
34.5	63.6	1.15e+00	0.798	0.06003		0.6505		0.945
35.5	62.4	3.01e+00	0.759	0.06209		0.5992		0.920
38.5	57.4	2.37e+00	0.728	0.06250		0.5598		0.896
39.5	55.0	4.40e+00	0.670	0.06421		0.4820		0.858
40.5	50.6	2.49e-04	0.670	0.06421		0.4820		0.858
41.5	50.6	2.38e-04	0.670	0.06421		0.4820		0.858
43.5	50.6	3.42e+00	0.625	0.06276		0.4277		0.822
44.5	46.2	4.94e-06	0.625	0.06276		0.4277		0.822
47.5	46.2	7.81e-01	0.614	0.06197		0.4162		0.812
48.5	45.4	8.42e+00	0.500	0.05542		0.2831		0.717
49.5	37.0	1.75e-05	0.500	0.05542		0.2831		0.717
51.5	37.0	5.84e-07	0.500	0.05542		0.2831		0.717
52.5	36.0	4.03e-04	0.500	0.05542		0.2831		0.717
56.5	35.0	6.13e+00	0.413	0.04402		0.2034		0.622
58.5	28.9	1.18e-05	0.413	0.04402		0.2034		0.622
60.5	26.9	4.70e+00	0.340	0.03380		0.1456		0.535
61.5	22.2	3.96e-07	0.340	0.03380		0.1456		0.535
62.5	22.2	4.11e+00	0.277	0.02450		0.1039		0.450
64.5	17.1	5.00e-02	0.276	0.02441		0.1033		0.449
66.5	16.0	2.14e-15	0.276	0.02441		0.1033		0.449
74.0	15.0	2.67e+00	0.227	0.01863		0.0665		0.388
76.0	12.3	1.33e+00	0.203	0.01604		0.0475		0.358
79.5	11.0	4.00e+00	0.129	0.00685		0.0249		0.233
80.5	6.0	5.07e-12	0.129	0.00685		0.0249		0.233
97.5	3.0	1.00e+00	0.086	0.00429		0.0000		0.184

After doing the necessary plots we get,

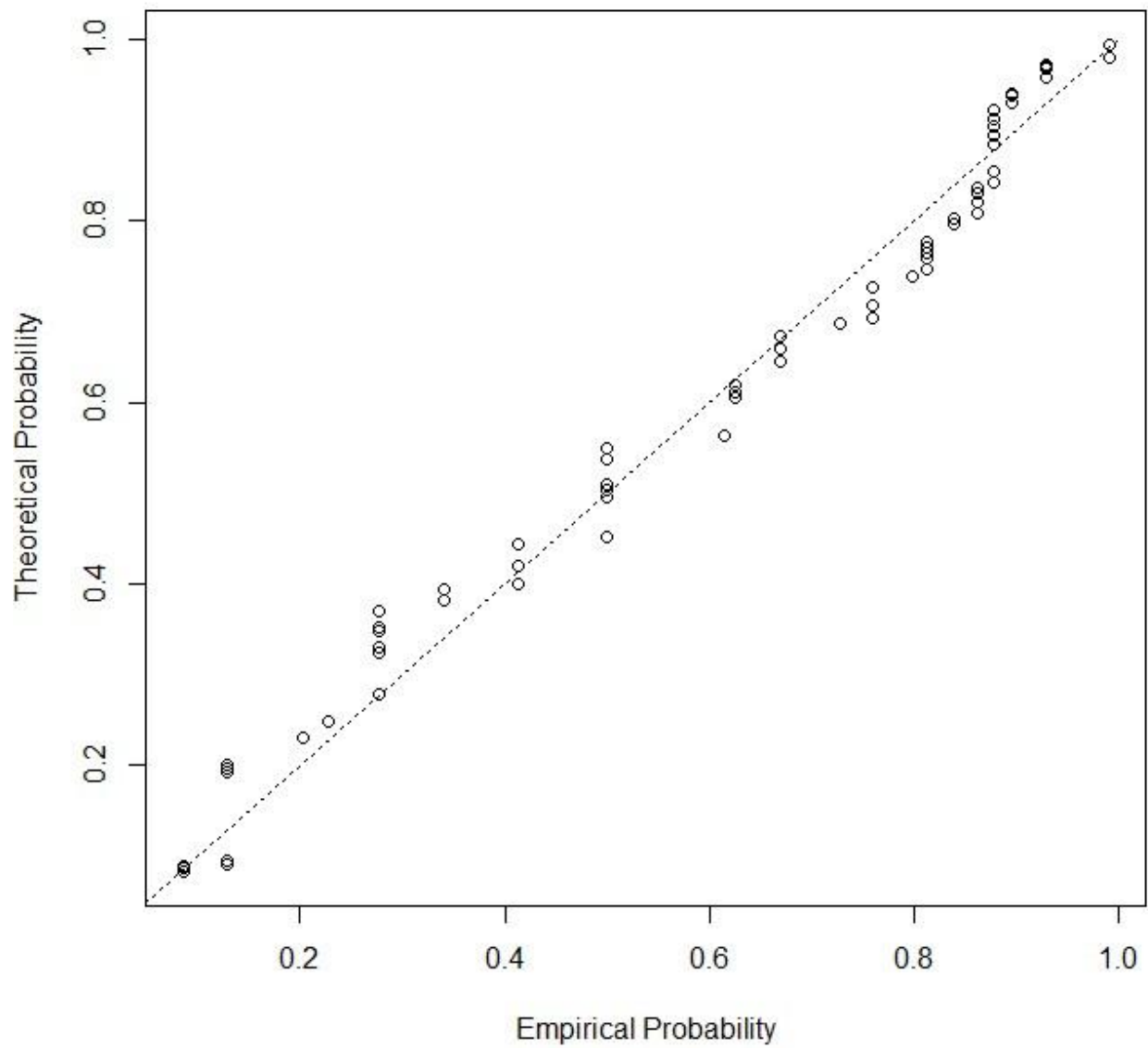
Comparison of parametric and non parametric survivor functions



Comment: After plotting the estimated parametric survivor function, then overlaying the nonparametric survivor function, we can say that the Weibull distribution looks like a good choice as a parametric model for the remission times for these patients.

Model Assessment Using P-P Plot

Weibull P-P Plot (No Adjustment)



Comment: From the plot we can conclude, Weibull model appears to be a reasonable choice.

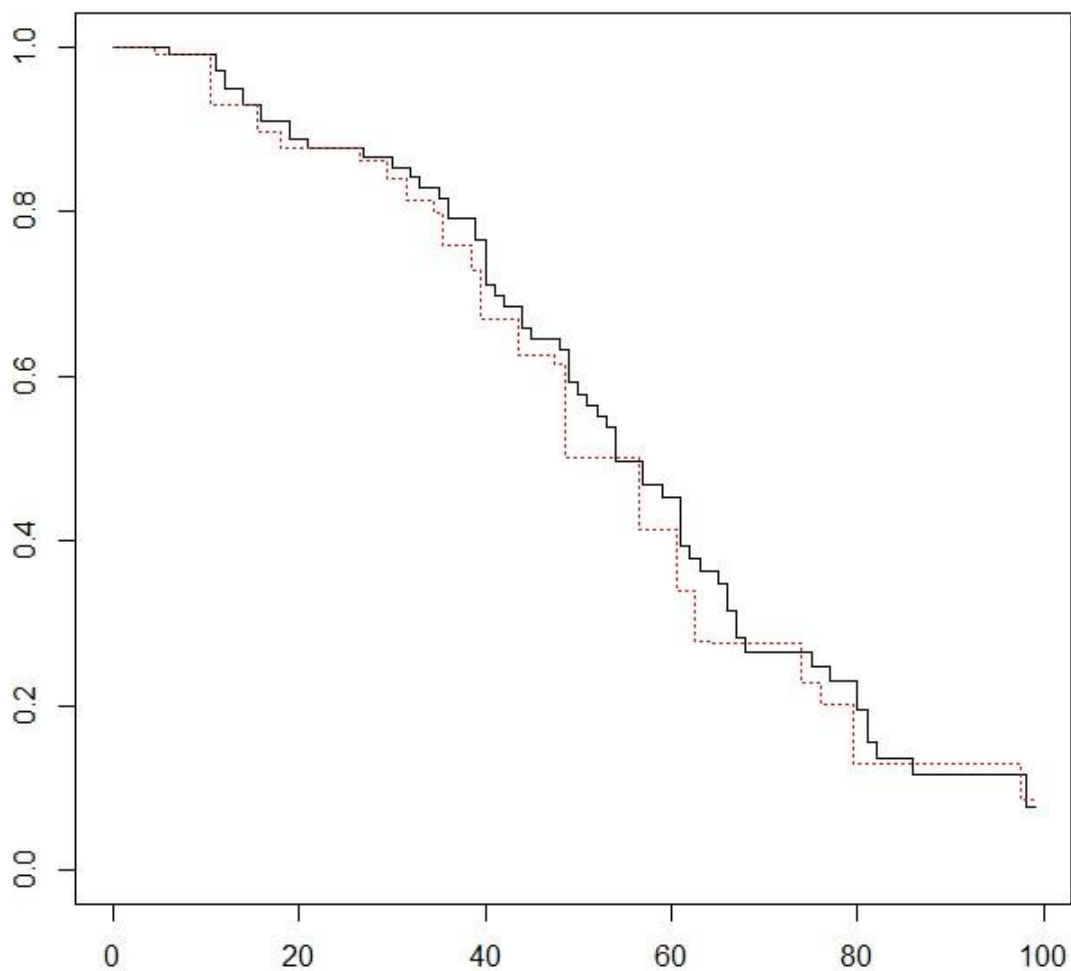
Comparison with Right-Censored Analysis

The Recoded data is given below,

```
[1] 33 44+ 30 80 61 28+ 35 19 22+ 45 29+ 52+ 54 63 57 38+ 54 36
[19] 32+ 50 44 51 57 26+ 17+ 97+ 99+ 61 49 77 40 56+ 80 62 67 97+
[37] 44 40 86 41 30+ 52 67 68 39 39 16+ 48 71+ 53 37+ 36 60+ 25+
[55] 21+ 54 16 11 61 21+ 75 60+ 49 49 40 14 12 81 59 33+ 42 66
[73] 65 19 64+ 9+ 96+ 98 11 66 21 6 80+ 66+ 82 61 98+ 12 17+ 32
[91] 16 34+ 11+ 14 19+ 40 81 27+ 27+ 27
```

After doing the necessary plots we get, (Didn't draw the confidence intervals for any one of them)

Comparison of non-parametric survivor functions in both cases



red line = considering interval censoring, black line = considering right censoring

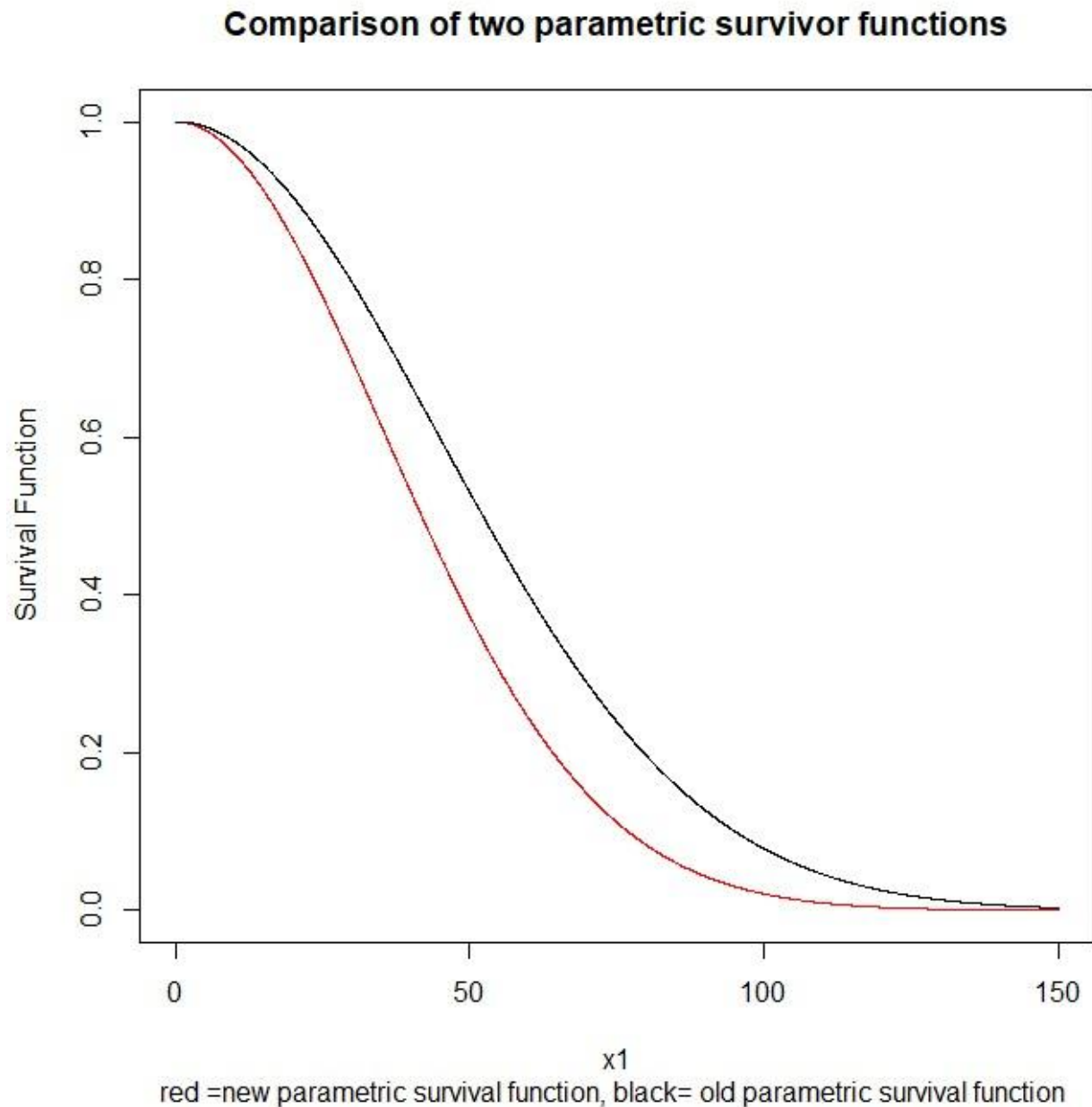
Comment: Both curves look similar. The difference doesn't look drastic. But, further investigation might be needed to determine if the difference matters or not.

The MLEs are calculated as,

Shape parameter = 1.971164

Scale parameter = 50.365

Now we plot a comparison of parametric survivor curves,



Comment: The two parametric survivor curves suggest that they are very different from one another.

Observation: Treating interval-censored observations as if the event occurred exactly at the right endpoint may result in overestimation of survival times. Since only the interval containing the event is known, the exact occurrence time cannot be determined from the available data.

Reduced Sample Analysis

With one-fourth of the data:

If we redo the code with one fourth of the data, we get,

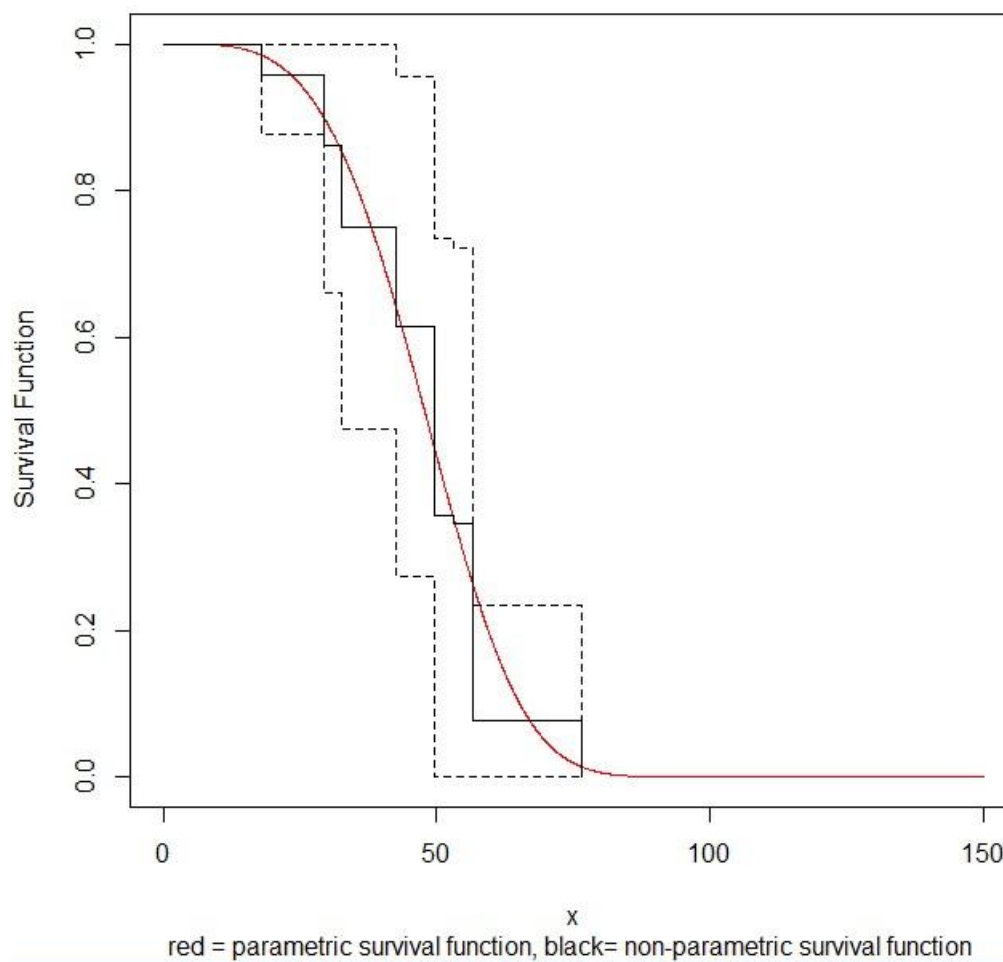
The MLEs are calculated as,

Shape parameter = 3.860547

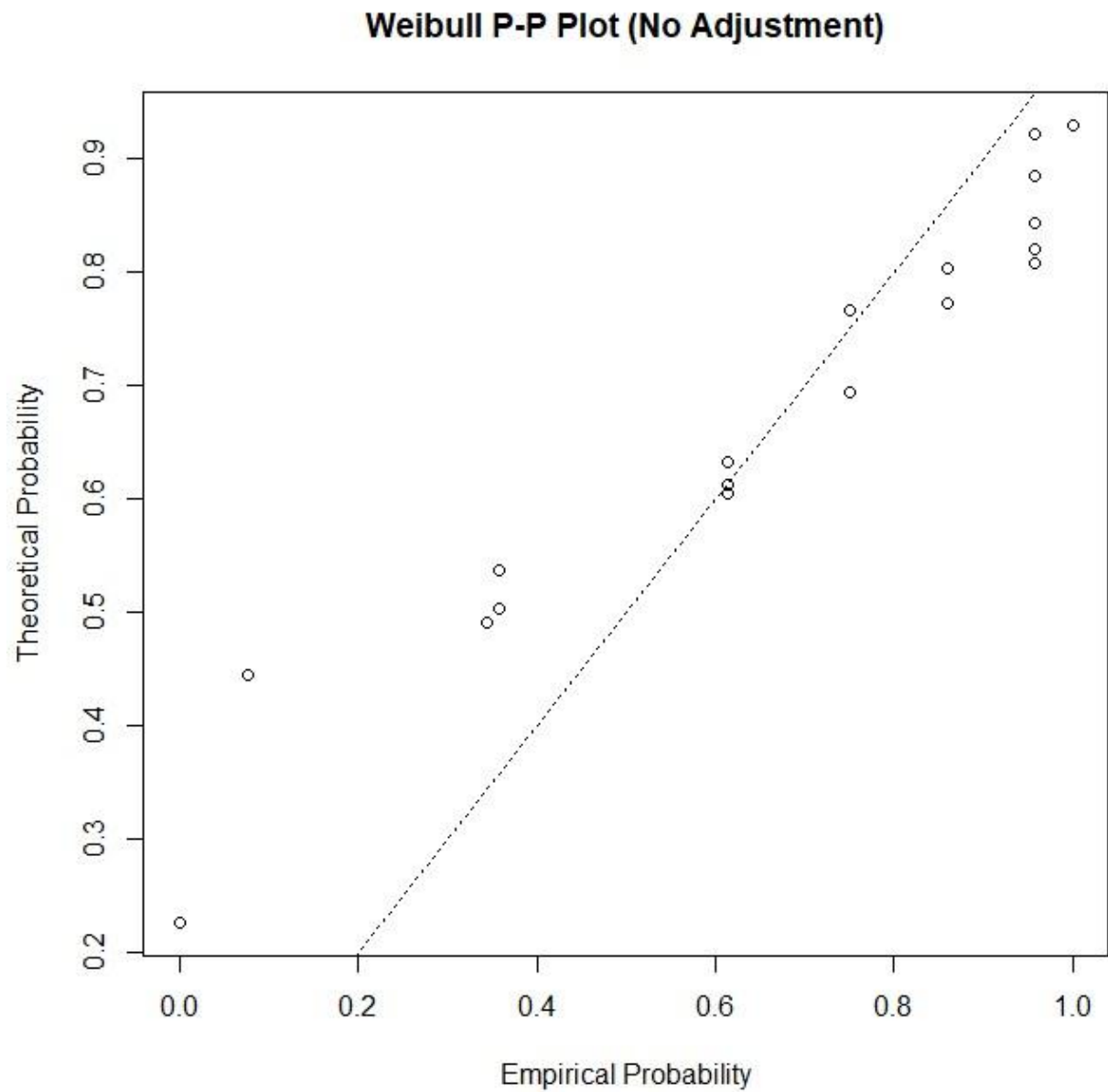
Scale parameter = 52.5391

The necessary plot,

Comparison of parametric and non parametric survivor functions



After plotting the estimated parametric survivor function, then overlaying the nonparametric survivor function, we can say that the Weibull distribution does not look like a viable choice as a parametric model for the remission times for these patients. The P-P plot,



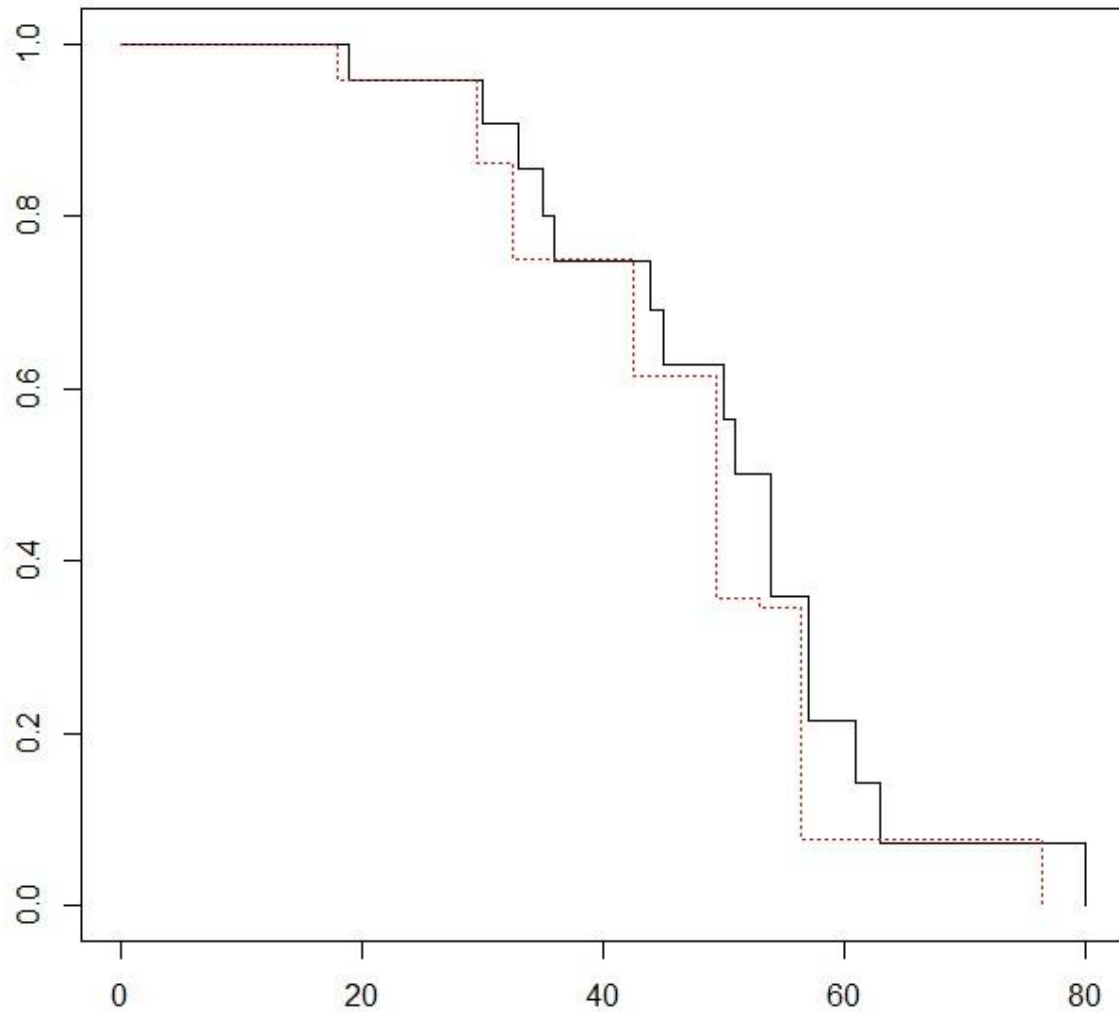
The P-P plot suggests that for a small sample, our considered model might not be so good.

After recoding the data to our preference we get the data as,

```
33 44+ 30 80 61 28+ 35 19 22+ 45 29+ 52+ 54 63 57 38+ 54 36 32+
50 44 51 57 26+ 17+
```


Now to observe the non-parametric survivor functions,

Comparison of non-parametric survivor functions in both cases



red line = considering interval censoring, black line = considering right censoring

There appears to be some difference between the two non-parametric survivor functions.

For a right censored case,

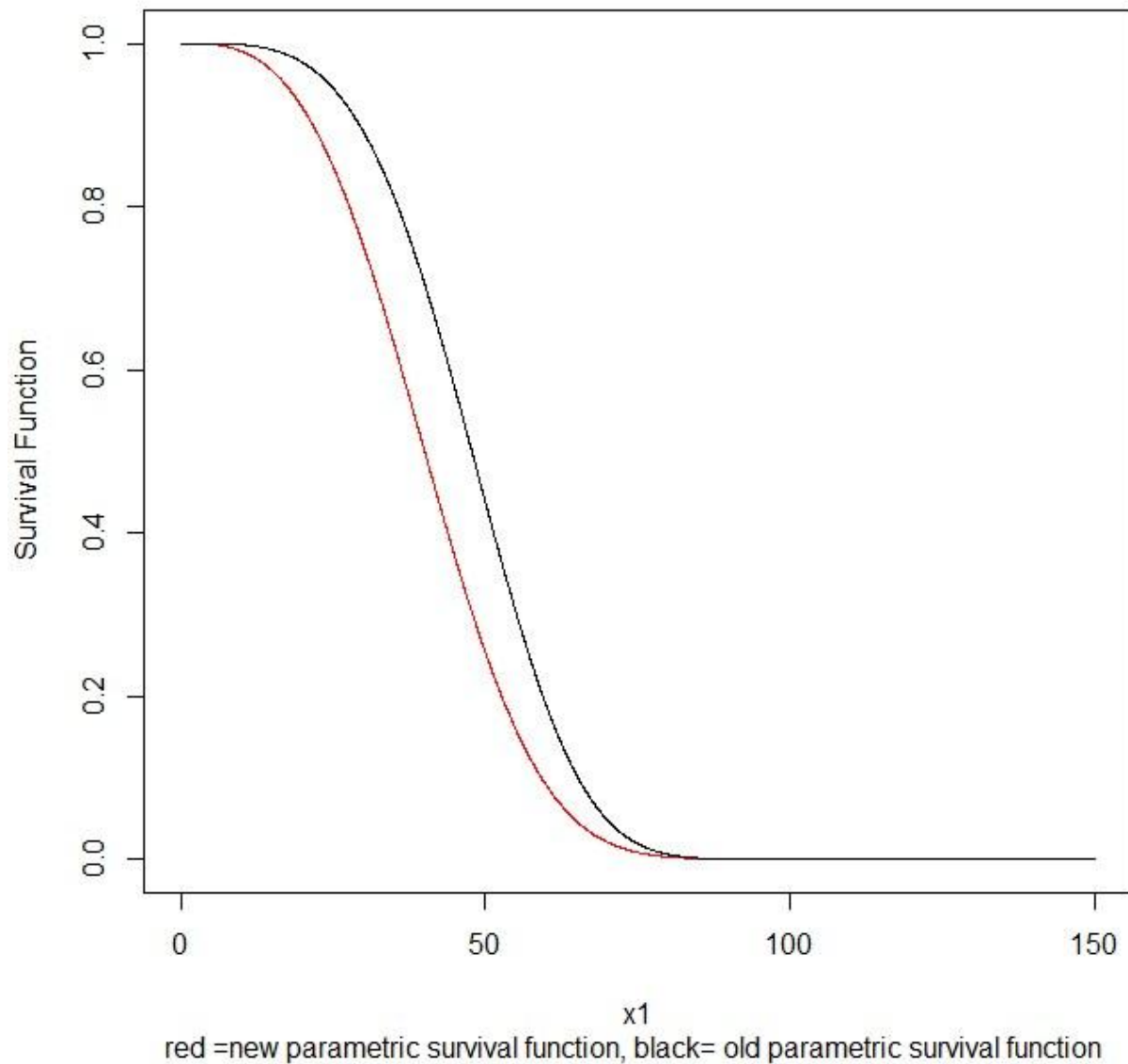
The MLEs are calculated as,

Shape parameter = 3.06551

Scale parameter = 45.09824

The necessary plot is given below,

Comparison of two parametric survivor functions



Comment: The two parametric survivor curves suggest that they are very different from one another. Even for a small sample.

Discussion

This project analyzed prostate cancer remission data using interval-censored survival analysis techniques. A Weibull distribution was proposed as a parametric model, and the resulting survivor function was compared with a nonparametric estimate. For the complete dataset, the Weibull model appeared to provide a reasonable fit, as the parametric and nonparametric survivor functions showed good agreement. The P-P plot also supported the suitability of the Weibull distribution for modeling remission times.

The analysis further explored the impact of treating interval-censored observations as right-censored observations. Although the nonparametric survivor functions did not appear drastically different, noticeable differences were

observed in the corresponding parametric survivor functions. This suggests that the method used to handle censoring can influence parameter estimates and the resulting survival curves.

When the analysis was repeated using only one-fourth of the available observations, the fit of the Weibull model deteriorated. The parametric and nonparametric survivor functions showed less agreement, and the P–P plot indicated that the Weibull model might not be an appropriate choice for the smaller sample. This demonstrates the effect of sample size on model assessment and parameter estimation.

Overall, the results highlight the importance of properly accounting for interval censoring in survival studies. Simplifying interval-censored observations by assigning exact event times may lead to different conclusions and potentially biased estimates of survival.

A Reasonable Alternative:

The “left” value of the intervals can be considered instead of the right values. Meaning, for an interval $(a, b]$, we will consider the time a , because we know for a fact that the participants have survived up to that time point. In this way, we will not be overestimating the survival times. We will be working with information that is known with certainty from the observed data.

This approach provides a simple alternative for researchers who prefer a right-censored analysis while avoiding the assumption that the event occurred exactly at the right endpoint of the interval. Although it does not fully utilize the interval-censored nature of the data, it is more consistent with the information that is actually observed.